

Cape Peninsula University of Technology

Department of Chemical Engineering
Investigative Projects (IVP360S) | Semester 1, 2026

Monitoring the Impact of Blending Borehole Water and Treated Wastewater on Dam Water Quality for Potential Reuse

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1. Introduction: Background and Motivation

The aim of this project is to evaluate the physical and chemical quality of a blended water dam supplied by borehole water and treated municipal wastewater, and to determine its suitability for potential reuse and other end-use purposes. As freshwater sources such as rivers and rain-fed dams become less reliable due to climate variability and growing urban demand, communities and institutions are increasingly turning to alternative water sources. Two of the most accessible alternatives are groundwater extracted from boreholes and treated municipal wastewater. At CPUT, both sources are pumped into a single recreational dam, creating a blended water body whose quality must be understood before any reuse application is considered.

Although both sources may appear suitable for storage individually, they differ substantially in chemical composition. Borehole water typically carries elevated dissolved minerals such as calcium, magnesium, and salts acquired through prolonged contact with subsurface geological formations. Treated wastewater, even after municipal processing, may retain residual nutrients including nitrogen and phosphorus compounds, as well as organic matter from human activity. When these chemically distinct waters blend in an open, sunlit dam, the resulting mixture can present challenges including accelerated algal growth driven by nutrient loading, mineral scaling on pipes and equipment, fluctuating pH, and elevated turbidity.

1.1 Objectives

- Characterise the source waters by testing and comparing the physical and chemical properties of the borehole water and treated wastewater before they enter the dam.
- Monitor the blending effect to determine how the mixing of the two sources affects overall water quality within the dam.
- Evaluate purification requirements and propose a treatment process suitable for the intended reuse application, using available data and supporting literature where experimental trials were not completed.

2. Literature Review

2.1 Borehole Water Quality

Groundwater quality is largely determined by the lithology of the aquifer through which it travels. Water that has percolated through granitic or sedimentary formations over extended periods tends to accumulate dissolved minerals including calcium, magnesium, bicarbonates, and silica through mineral dissolution and ion exchange processes. (MACKENZIE L. DAVIS, 2013)note that while larger cities are typically supplied by surface water, smaller communities

and institutions more frequently rely on groundwater from wells and boreholes. Groundwater benefits from natural filtration as it passes through soil and rock layers, but this same process results in the dissolution of minerals into the water, making its chemical composition dependent on the geology of the aquifer.

In the Cape Flats area, where CPUT is situated, groundwater monitoring has recorded TDS values ranging from 260 to 1 600 mg/L across different borehole locations, with chemical types characterised primarily by calcium-bicarbonate and sodium-calcium-chloride compositions (Segun Adelana, 2010). The measured TDS of 362–369 mg/L in this investigation falls within the lower range reported for the Cape Flats aquifer, consistent with a relatively uncontaminated borehole source. The dominance of calcium-bicarbonate chemistry in Cape Flats groundwater accounts directly for the elevated pH observed in this study, as dissolved bicarbonates raise the alkalinity and buffer the water toward alkaline conditions. (Segun Adelana, 2010)

2.2 Treated Municipal Wastewater

Municipal wastewater undergoes primary, secondary, and in some cases tertiary treatment before discharge or reuse. Secondary biological treatment removes a significant portion of organic load measured as chemical oxygen demand (COD) or biological oxygen demand (BOD), but nutrient removal, specifically of nitrogen and phosphorus, requires additional treatment stages that are not always fully implemented at all treatment works.

The Water Research Commission (WRC) documents common secondary treatment configurations used in South Africa, including trickling filters, rotating biological contactors, and activated sludge processes. The activated sludge process is the most widely applied, capable of achieving significant COD and nitrogen removal through nitrification and denitrification. Where nutrient removal is required, biological nutrient removal (BNR) processes such as the 3-stage Bardenpho or UCT process are employed. However, where such advanced treatment is absent, the treated effluent may still contain residual nitrates and phosphates that present a risk of nutrient loading when discharged into a receiving water body (Water Research Commission, N/A) (Segun Adelana, 2010)

2.3 Blended Water Systems and Associated Risks

When groundwater and treated wastewater are blended, the resulting mixture can exhibit intermediate chemical characteristics that present unique management challenges. Elevated nutrient levels from the wastewater fraction may promote eutrophication in the dam, particularly when combined with warm temperatures and high sunlight exposure. Eutrophication leads to excessive algal growth, oxygen depletion in lower water layers, and potential production of harmful algal toxins (Boyd, 2015).

Simultaneously, the mineral content introduced by the borehole water may contribute to water hardness and scaling potential. The Langelier Saturation Index (LSI) is commonly used to assess whether a blended water is likely to deposit calcium carbonate scale or be corrosive to pipe surfaces. A positive LSI indicates scale-forming tendency, while a negative index suggests corrosive conditions. Without hardness data from this investigation, the LSI cannot be calculated, and this is noted as a limitation.

(Boyd, 2015)

2.4 South African Water Quality Guidelines

The Department of Water and Sanitation (DWS) publishes water quality guidelines covering domestic, recreational, industrial, irrigation, livestock, aquaculture, and aquatic ecosystem uses. The South African National Standard SANS 241 (2015) sets operational and absolute limits for drinking water parameters including pH, turbidity, TDS, conductivity, and microbiological indicators. For this investigation, the blended dam water is most appropriately benchmarked against DWAF Volume 2 (Recreational Use) and SANS 241 as upper

references, given that the intended end use is industrial cooling water, which sits below potable standards but must meet basic stability and equipment-protection thresholds (DWAF, 1996b; SANS, 2015; Kruger, 2022). (South African National Standards)

Key guideline values used in this investigation are pH 6.5–8.5, turbidity ≤ 10 NTU for recreational and ≤ 1 NTU for potable use, and TDS ≤ 450 mg/L for domestic use. For industrial cooling water applications, additional parameters of relevance include hardness (to assess scaling risk) and conductivity (to assess corrosion potential), neither of which was measured in this investigation. (South African National Standards)

2.5 Treatment Options for Blended Water

Where blended water does not meet guideline requirements, treatment is necessary before reuse. Two treatment processes are proposed based on the water quality profile identified in this investigation: coagulation-flocculation with sedimentation for turbidity removal, and pH correction for alkalinity control.

Coagulation-flocculation involves the addition of a chemical coagulant, typically aluminium sulfate (alum, $\text{Al}_2(\text{SO}_4)_3$), to destabilise suspended particles and colloidal matter in the water. Under rapid mixing, the coagulant neutralises the surface charge of particles, allowing them to aggregate. Gentle slow mixing in a flocculation chamber promotes growth of larger settleable flocs, which are then removed by gravity sedimentation. A jar test is the standard laboratory procedure used to determine the optimal coagulant dose for a specific water sample (MACKENZIE L. DAVIS, 2013)

pH correction for an alkaline water can be achieved by controlled dosing of carbon dioxide, which reacts with hydroxide and bicarbonate ions to lower pH toward a neutral range. CO_2 dosing is preferred over mineral acid dosing in water treatment applications as it does not introduce additional ionic species and is precisely controllable (MACKENZIE L. DAVIS, 2013). The target pH range for the treated water is 6.5 to 8.5 in accordance with SANS 241. (South African National Standards)

3. Research Methodology and Discussion of Results

The investigation was conducted in five stages aligned with the standard workflow of a water quality technician: site assessment and field observations, standardised water sampling, laboratory physical and chemical analysis, purification and treatability trials, and data comparison with applicable guidelines. The results obtained from each stage were presented and discussed in relation to the project objectives.

3.1 Site Assessment and Field Observations

A site assessment was conducted at the CPUT Bellville Campus dam on 10 April 2026 between 09:30 and 10:00. Weather conditions at the time were moderately sunny with strong wind. Visual inspection of the dam revealed dark, murky water of a deep green colouration, indicative of suspended particulate matter and biological activity. Visible algal growth was observed at the water surface, suggesting active photosynthetic activity consistent with nutrient loading from the mixed inflow sources. Whether this represents early-stage eutrophication or general bacterial and algal colonisation could not be confirmed without microbiological testing; however, the visual evidence is consistent with elevated nutrient concentrations expected from the treated wastewater fraction. The dam infrastructure, including visible pipework at the inflow points, appeared to be in good condition with no visible damage or corrosion. Wildlife, specifically a resident duck population, was observed inhabiting the dam area, indicating that the water body functions as a living ecosystem and that any treatment or reuse strategy must consider ecological impact.



Figure 1 water sources

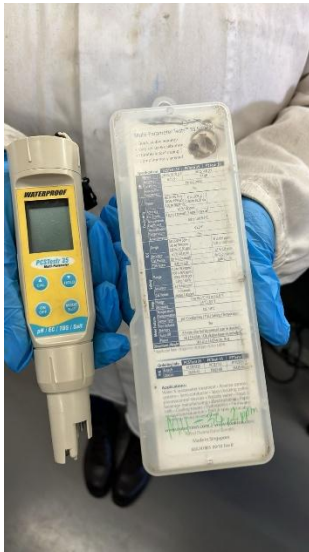
3.2 Standardised Water Sampling

Water samples were collected at the CPUT Bellville Campus dam on 29 April 2026 at 10:30. Samples were drawn from the designated sampling points in the pool area of the campus. One-litre samples were collected in labelled bottles and stored in a cooler box immediately following collection to prevent parameter degradation during transport to the laboratory.



Figure 2 water collection

3.3 Laboratory



Physical and Chemical Analysis



Figure 3 Multi Parameter tesr 35 Series used to measure ppm (300ppm standard)

Figure 4 standard solution (300ppm)



Figure 6 50 ml sample



Figure 5 cleaning with de-ionized water



Figure 7 conductivity standard 1288-uS/cm



Figure 8 conductivity recording



Figure 9 Turbid master



Figure 10 Turbidity standards, 0.02 NTU, 10 NTU, and 1 000 NTU



Figure 11 PH testing

Laboratory analysis was performed on the collected samples to measure physical and chemical parameters. The pH meter was calibrated using buffer solutions at pH 4.01 and 7.00. The conductivity meter was calibrated with a 12 880 $\mu\text{S}/\text{cm}$ standard solution. Turbidity was measured using a turbidimeter calibrated against three standard solutions (0.02 NTU, 10 NTU, and 1 000 NTU). Total dissolved solids were measured using a TDS meter calibrated at 300 ppm. Results are presented in Table 1 below.

Table 1: Measured water quality parameters for the mixed dam sample compared with SANS 241 and DWAF recreational guidelines

Parameter	Mixed Sample	SANS 241 Recreational	SANS 241 Domestic	Status
pH	8.08 – 8.79	6.5 – 8.5	6.5 – 8.5	Too high
Temperature ($^{\circ}\text{C}$)	18.4 – 19.4	No limit	No limit	✓ Acceptable

Parameter	Mixed Sample	SANS 241 Recreational	SANS 241 Domestic	Status
Conductivity ($\mu\text{S/cm}$)	82.8 – 89.0	< 1 500	< 1 500	✓ Well within
Turbidity (NTU)	5.41 – 10.59	≤ 10	≤ 1	⚠ Marginal
TDS (mg/L)	362 – 369	$\leq 1\,000$	≤ 450	✓ Within range
Total Hardness	Not measured	20 – 100 mg/L	–	— Data gap
Nitrates	Not measured	< 6 000 $\mu\text{g/L}$	$\leq 11\,000\ \mu\text{g/L}$	— Data gap
Phosphates	Not measured	< 500 $\mu\text{g/L}$	–	— Data gap
COD	Not measured	–	–	— Data gap

Note: BH1 (CPUT treated water) data was unavailable at time of reporting due to data collection scheduling constraints within the group allocation. The values presented reflect the mixed dam sample as the primary dataset, supplemented by BH2 (raw municipal water) readings from a parallel group. Hardness, nitrates, phosphates, and COD were not measured and are identified as data gaps in Section 7.

3.4 Purification and Treatability Trials

Jar testing and chlorine demand testing were not completed during this investigation due to logistical constraints in the laboratory schedule and gaps in task allocation within the group. These limitations are acknowledged in full in Section 7 (Limitations). In the absence of experimental treatability data, the proposed design in Section 4 is based on literature-derived dosing parameters and the water quality profile established in Table 1.

3.5 Discussion of Results

The measured pH of the mixed dam sample ranged from 8.08 to 8.79. The upper value of 8.79 exceeds the SANS 241 limit of 8.5 for domestic use and approaches the ceiling of the recreational use guideline. Elevated pH in a dam receiving borehole water is consistent with the calcium-bicarbonate chemistry characteristic of the Cape Flats aquifer, as described by Xu et al. (2007). Dissolved bicarbonates from groundwater transit through calcareous formations raise alkalinity and buffer the water toward alkaline conditions. The treated wastewater fraction may additionally contribute residual alkalinity from biological treatment processes.

Turbidity ranged from 5.41 to 10.59 NTU. The lower value falls within the SANS 241 recreational limit of 10 NTU, while the upper value approaches this ceiling. For industrial reuse — particularly cooling water applications — turbidity should ideally be as low as possible to minimise fouling of heat exchangers and distribution pipework. The source of turbidity is likely suspended particulate matter from both inflows, potentially including fine algal material if biological growth has commenced in the dam.

Conductivity at 82.8 to 89.0 $\mu\text{S/cm}$ is well within all guideline limits, indicating that the total dissolved ion load is low. TDS at 362 to 369 mg/L is within the SANS 241 domestic limit of 450 mg/L (South African National Standards), suggesting no immediate desalination requirement. However, without hardness data, the scaling potential of the water for industrial cooling systems cannot be quantitatively assessed. (South African National Standards)

The absence of hardness, nitrate, phosphate, and COD data limits the completeness of this assessment. Given that BH2 is a municipal water source, residual nutrient loading is probable, and eutrophication risk within the dam cannot be excluded without further testing.

4. Proposed Design

Based on the water quality profile in Table 1, the intended end-use application for the treated blended dam water is industrial cooling water. This application requires water that is free of suspended solids that would foul heat exchangers, stable in pH to prevent corrosion or scaling, and low in dissolved minerals that could deposit on heat transfer surfaces. The two parameters that deviate from target values are pH (maximum 8.79, exceeding the SANS 241 limit of 8.5) (South African National Standards) and turbidity (up to 10.59 NTU, exceeding the practical threshold for cooling water applications). A two-stage treatment process is proposed to address these parameters.

4.1 Stage 1: Coagulation-Flocculation and Sedimentation (Turbidity Removal)

Coagulation-flocculation followed by gravity sedimentation is the standard first stage in treating turbid surface or blended water. Aluminium sulfate (alum, $\text{Al}_2(\text{SO}_4)_3$) is dosed into the water under rapid mixing to destabilise colloidal particles by charge neutralisation. Gentle slow mixing (flocculation) in a dedicated chamber then promotes aggregation of the destabilised particles into larger, settleable flocs. These flocs settle under gravity in a sedimentation basin, and the clarified water overflows from the top while sludge accumulates at the base for periodic removal. (MACKENZIE L. DAVIS, 2013)

The optimal coagulant dose for this water would normally be determined experimentally by a jar test using the actual sample. Since no jar test was conducted in this investigation, a preliminary dose range of 5–30 mg/L alum is indicated in the literature for water of similar turbidity (5–15 NTU) (MACKENZIE L. DAVIS, 2013). This range should be confirmed by a jar test before implementation. The sedimentation basin should be sized to provide a surface overflow rate consistent with the design flow and the settling velocity of the expected flocs, typically 1–2 $\text{m}^3/\text{m}^2\cdot\text{h}$ for water of this type.

4.2 Stage 2: pH Correction by CO_2 Dosing

Following clarification, the elevated pH of up to 8.79 is addressed by controlled CO_2 dosing. Carbon dioxide dissolves in water to form carbonic acid, which reacts with bicarbonate and hydroxide ions to lower pH toward the neutral range. The target outlet pH for the treated water is 6.5 to 8.5 in accordance with SANS 241, with a preferred operating target of 7.0 to 7.5 for cooling water applications to minimise both corrosion and scaling risk.

CO_2 dosing is preferred over mineral acid dosing at this scale as it does not introduce additional anions such as chloride or sulfate, which could accelerate corrosion in cooling systems. Dosing is applied downstream of the sedimentation stage, treating clarified water before final storage. A pH monitoring point is included at the outlet of the pH correction unit to enable real-time feedback control of the CO_2 dose (Twort, 2000)

4.3 Process Flow Description

The full process flow is illustrated in the process flow diagram included in Section 6. The sequence of unit operations is as follows:

- Dam water is drawn from the blended dam and conveyed by pump to the rapid mix coagulation tank.
- Alum is dosed into the rapid mix tank at 5–30 mg/L. Rapid mixing ensures adequate coagulant dispersion and particle destabilisation.

- Water flows from the coagulation tank into the flocculation chamber where gentle agitation promotes floc growth.
- Flocculated water enters the sedimentation basin. Settled sludge is periodically removed from the base. Clarified water exits from the top.
- A turbidity meter at the clarified water outlet confirms turbidity removal before the next stage.
- Clarified water passes through the CO₂ dosing point where pH is adjusted to the target range of 6.5–8.5. A pH monitor confirms outlet quality.
- Treated water is discharged to a storage reservoir for use as industrial cooling water.

5. Design Evaluation Using Technical and Economic Indices

5.1 Technical Evaluation

The proposed two-stage process of coagulation-sedimentation followed by CO₂ pH correction is technically appropriate for the identified water quality issues. Table 2 provides a qualitative comparison of the selected processes against alternatives considered.

Table 2: Qualitative technical comparison of treatment options

Criterion	Coagulation-Sedimentation	Sand Filtration (alternative)	CO ₂ pH Correction
Target parameter	Turbidity, suspended solids	Turbidity, fine particles	Elevated pH / alkalinity
Effectiveness	High for colloidal turbidity	Moderate; misses colloidal particles without pre-coagulation	High; precise and reversible
Chemical required	Alum (Al ₂ (SO ₄) ₃)	None	CO ₂ gas or dilute acid
Operational complexity	Low	Low; requires backwash	Low; requires dosing control
Capital cost (relative)	Low–moderate	Low	Low
Selected for design	Yes	Not selected (requires pre-coagulation to match effectiveness)	Yes

Coagulation-sedimentation was selected over sand filtration alone because the turbidity in this water is partly colloidal in nature, and sand filtration without pre-coagulation is less effective at removing colloidal particles. Coagulation-sedimentation followed by sand filtration (polishing) would be the full conventional treatment train, but given the relatively moderate turbidity of this water and the absence of potable reuse requirements, the sedimentation stage alone is considered sufficient for the industrial cooling water application.

CO₂ dosing was selected over mineral acid addition because it avoids the introduction of additional anions, is more easily controlled to a target pH, and is less hazardous to handle at small scale.

5.2 Economic Evaluation

A full quantitative economic evaluation was not possible within the scope of this investigation, as no flow rate data was collected for the dam inflows, and equipment sizing therefore could not be completed. The following qualitative observations are provided based on available literature.

Both proposed unit processes — coagulation-sedimentation and CO₂ pH correction — are among the lowest-cost water treatment options available. Alum is widely available in South Africa at low cost per kilogram, and at a dose of 5–30 mg/L the chemical cost per cubic metre of treated water is low. CO₂ supply for a small-scale system is typically provided by standard pressurised cylinders, which are low-capital and low-maintenance. Neither process requires significant energy input beyond pumping.

The primary economic constraint identified is the cost of quantifying the jar test dose experimentally and commissioning the physical infrastructure. These costs were beyond the scope of the current investigation. A detailed cost analysis would require knowledge of the design flow rate, which should be obtained from the dam operators as a prerequisite for detailed design.

6. Final Design Including Process Flow Diagram

The final selected design is a two-stage treatment system comprising coagulation-flocculation with gravity sedimentation (Stage 1) followed by CO₂ pH correction (Stage 2). This system is designed to bring the blended dam water within SANS 241 and DWAF guidelines for turbidity and pH, and to render it suitable for use as industrial cooling water.

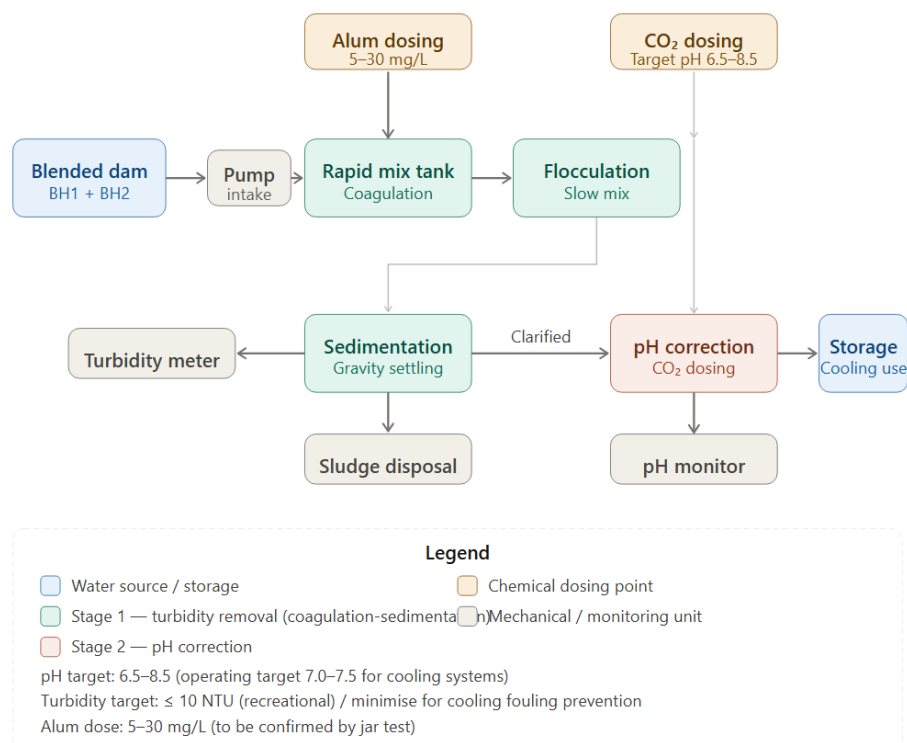


Figure 12 process flow diagram

The diagram shows the following key features:

- Chemical dosing points are clearly identified for both alum (Stage 1 inlet) and CO₂ (Stage 2 inlet).
- Monitoring points are provided at the sedimentation outlet (turbidity) and the pH correction outlet (pH), enabling quality control before the treated water enters storage.
- A sludge disposal stream exits the base of the sedimentation basin.
- The intake pump provides the hydraulic head to convey water through the treatment train by gravity from the coagulation tank onward.

For a formal P&ID, the following instrumentation and control elements would additionally be shown: flow control valve on the alum dosing pump, pH transmitter and controller on the CO₂ dosing line, turbidimeter with alarm on the sedimentation outlet, and a level indicator on the storage tank. These elements are described here in lieu of a fully drafted P&ID, which would require process flow rates and equipment sizing data not available within the scope of this investigation.

7. Conclusion

This investigation evaluated the physical and chemical quality of a blended dam water system supplied by borehole water and treated municipal wastewater at CPUT, with the aim of determining its suitability for reuse as industrial cooling water. Based on the parameters measured, the blended water does not currently meet all requirements for the intended application, primarily due to elevated pH (up to 8.79, exceeding the SANS 241 limit of 8.5) and turbidity that presents a fouling risk for cooling system infrastructure (up to 10.59 NTU).

Conductivity and TDS values are well within guideline limits, and the overall dissolved ion load is low, indicating that the water does not require desalination. The elevated pH is consistent with the calcium-bicarbonate chemistry of the Cape Flats aquifer, as documented in the regional groundwater literature, and is a predictable characteristic of the borehole water source.

A two-stage treatment process comprising coagulation-flocculation with gravity sedimentation followed by CO₂ pH correction is proposed as the minimum viable intervention to render the blended water suitable for industrial cooling reuse. This design is technically appropriate, operationally simple, and low-cost relative to more advanced treatment alternatives.

It is recommended that: (1) a jar test be conducted to determine the optimal alum dose for this specific water; (2) the missing parameters (hardness, nitrates, phosphates, COD) be measured in a follow-up sampling campaign to complete the water quality characterisation; (3) a hardness-based Langelier Saturation Index calculation be performed to quantify scaling risk in the cooling system; and (4) the blending ratio of BH1 to BH2 inflows be documented to enable mass-balance assessment of blending effects.

7.1 Limitations

- BH1 (CPUT treated water) data was unavailable at time of reporting due to scheduling constraints in the group data collection process. The investigation therefore characterises the mixed sample and BH2 only.
- Jar testing and chlorine demand testing were not completed, limiting the experimental basis for the treatability and design sections. Design dosing parameters are based on literature values and require experimental confirmation.
- Total hardness, nitrates, phosphates, and COD were not measured during the laboratory phase, leaving key parameters for industrial reuse and eutrophication risk assessment uncharacterised.
- The blending ratio of BH1 to BH2 inflows to the dam is unknown, preventing a mass-balance analysis of blending effects on water quality.

- No flow rate data was collected from the dam, preventing quantitative equipment sizing and a detailed economic evaluation.
- (Twort, 2000)

8. References

- Boyd, C. E. (2015). *Water Quality : an introduction*. New York: Springer .
- MACKENZIE L. DAVIS, D. A. (2013). *Intorduction To Environmental Engineering*. New York: McGraw Hill.
- Segun Adelana, Y. X. (2010). A conceptual model for the development and management of the Cape Flats Aquifer. *Ground water Pollution in Africa*, 461-474.
- South African National Standards. (n.d.). *South African National Standards*. Retrieved from <https://repository.up.ac.za/server/api/core/bitstreams/aef48a44-229e-4c70-9d33-45c49a85f15d/content>
- Twort, A. R. (2000). *Water supply*. London: Arnold Publishers.
- Water Reasearch Commision. (N/A). *Wastewater Treatment Technologies : A guide*. Cape Town: SALGA.
- Department of Water Affairs and Forestry (DWAF). (1996a). South African Water Quality Guidelines. Volume 1: Domestic Use. 2nd ed. Pretoria: DWAF.
- Department of Water Affairs and Forestry (DWAF). (1996b). South African Water Quality Guidelines. Volume 2: Recreational Use. 2nd ed. Pretoria: DWAF.
- Department of Water Affairs and Forestry (DWAF). (1996c). South African Water Quality Guidelines. Volume 3: Industrial Use. 2nd ed. Pretoria: DWAF.
- South African National Standards (SANS). (2015). SANS 241-1:2015 Drinking Water. Part 1: Microbiological, Physical, Aesthetic and Chemical Determinands. 2nd ed. Pretoria: SABS.

UBRICS FOR GA 4: Demonstrate competence to conduct investigations of broadly defined Engineering problems using research methods, including research- based knowledge, design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.

Focus Areas	INDICATORS	Level 4 <i>Strong</i>			Level 3 <i>Competent</i>			Level 2 <i>Developing</i>		Level 1 <i>Needs Work</i>			Achieved
Examination and Evaluation.	Identifying Problems for Investigation: Ability to identify problems/ issues / topics for investigation.	Identifies relevant problems/issues/topics with minimal or no orientation, and is able to select those that are pertinent/ critical for investigation.			Identifies relevant problems/ issues/ topics with minimal orientation, that require investigation.			Some ability to identify problems/ issues/ topics that may or may not require investigation.		Minimal or no ability to identify problems/ issues/ topic that may or may not require investigation.			
		10	9	8	7	6	5	4	3	2	1	0	
	Collection of Background information: Ability to gather background information (existing knowledge, research, and/ or indications of the problem)	Collects sufficient relevant background information from appropriate sources, and is able to identify pertinent/ critical information.			Collects sufficient relevant background information from appropriate sources.			Collects some relevant background information from appropriate sources.		Minimal or no ability to collect relevant background information.			
		10	9	8	7	6	5	4	3	2	1	0	
	Procedure: Ability to <i>select</i> OR ability to <i>develop</i> a methodology or theoretical framework to investigate a problem.	Selects/ develops an optimal methodology or theoretical framework.			Selects/ develops an appropriate methodology or theoretical framework.			Selects/ develops a methodology or theoretical framework that is somewhat		Minimal or no ability to select/ develop a methodology or theoretical framework.			
		10	9	8	7	6	5	4	3	2	1	0	
	For experimental procedures, see Experiential Methods Rubric (4.1)												
	Organizing Evidence: Ability to organize evidence to demonstrate patterns, and highlight differences and/ or similarities.	Organizes evidence with perception, demonstrating patterns, and highlighting differences and/ or similarities integral to the investigation.			Organizes evidence to demonstrates patterns, differences and/ or similarities important to the investigation.			Organizes evidence in a way that is somewhat effective in demonstrating patterns, differences, and/ or		Minimal or no ability to organize evidence.			
		10	9	8	7	6	5	4	3	2	1	0	

	Conclusion and Recommendations: Ability to state conclusion and make recommendations as a result of the investigation.	States logical conclusion and makes insightful recommendations, and identifies those that are pertinent/ critical.	States logical conclusions and makes appropriate recommendations.			States conclusions and makes recommendation that may or may not be relevant to the investigation.		Minimal or no ability to state conclusions or make recommendations.			
		1098	7	6	5	4	3	2	1	0	
	Limitations and Implications: Ability to identify limitations and implications.	Identifies all significant limitations and implications.	Identifies many important limitations and implications.			Identifies some limitations and implications.		Minimal or no ability to identify limitations or implications.			
		1098	7	6	5	4	3	2	1	0	